

MIMIC-IV ICU 真实世界证据分析汇报

早期升压药使用与院内死亡风险： Target Trial Emulation + IPTW/AIPW

汇报目的

向老板和医学团队展示 RWE 项目的研究设计、数据提取、统计分析和医学解释。

当前结果为 preliminary analysis ，适合内部讨论和简历项目展示。

重点不是证明疗效，而是展示如何处理 ICU 观察性治疗策略中的 confounding by indication 。

9,502

Analytic cohort

3,772

Early VP 0-6h

5,730

No early VP

1.17 pp

AIPW risk difference

1. 研究问题与临床背景

ICU 治疗策略分析中，粗比较通常会受到疾病严重程度影响

Research Question

Adult sepsis-coded first ICU stays 中，ICU admission 后 6 小时内早期使用升压药是否与院内死亡风险相关？

Treatment strategy: early vasopressor initiation within 0-6 hours.

Comparator: no vasopressor initiation within 0-6 hours.

Why RWE Methods Matter

早期升压药患者通常更重，粗死亡率比较容易误导。

需要 target trial emulation 明确定义 time zero、grace period、baseline confounders。

使用 PS/IPTW/AIPW 评估 measured confounding 调整后的 treatment strategy association。

2. Target Trial Emulation 框架

将 EHR 观察性数据尽量整理成“模拟试验”的分析结构

Population

Adult first ICU stay + sepsis ICD proxy + 6h landmark

Treatment

Vasopressor started within 0-6h after ICU admission

Comparator

No vasopressor initiation within 0-6h

Outcome

Hospital mortality

Confounders

Demographics, admission type, care unit, first-6h vitals/labs

Estimand

Average effect / adjusted risk difference in target population

3. 数据构建流程

从源数据表构建 one-row-per-ICU-stay analysis dataset

Core Source Tables

hosp: patients, admissions, diagnoses_icd, labevents

icu: icustays, inputevents, d_items, chartevents

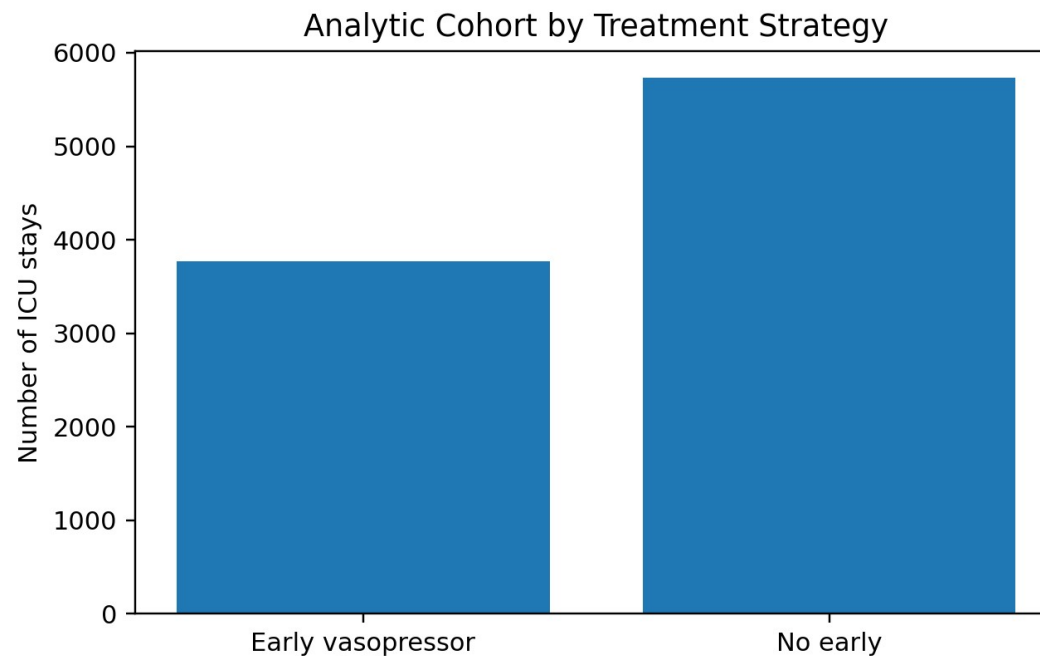
Output: patient/stay-level cohort + exposure + covariates + outcome

Derived Variables

first_vasopressor_time and early_vasopressor flag

first-6-hour vitals/labs: HR, MAP/MBP, RR, temperature, lactate, creatinine, WBC, platelet, albumin, bilirubin

hospital_expire_flag as primary outcome



4. Cohort Summary

当前可用于内部讨论和简历项目展示的 preliminary cohort

9,502

Analytic cohort

3,772

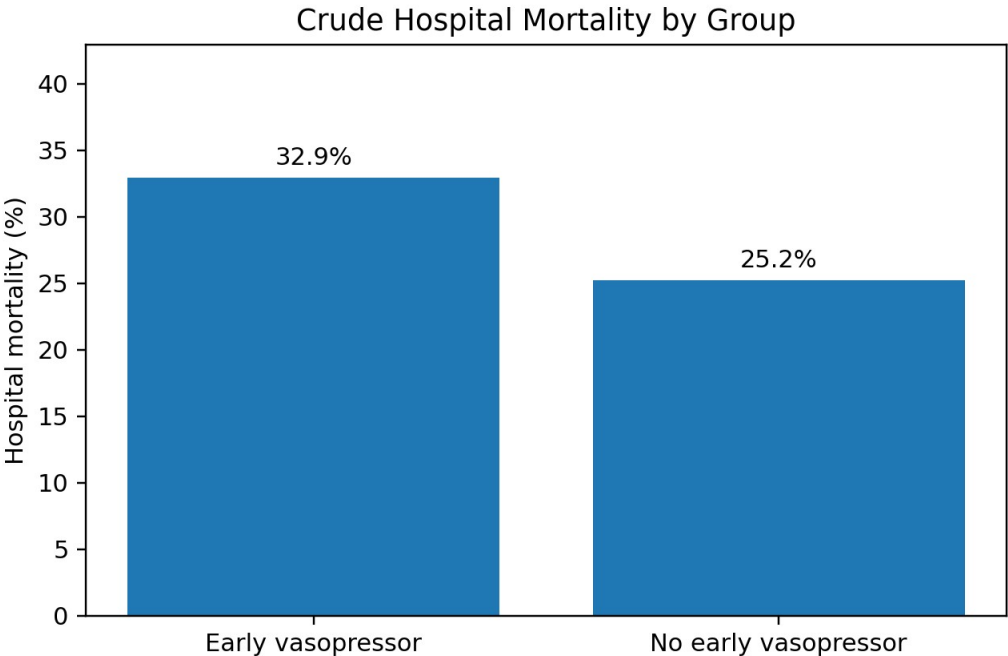
Early VP (39.7%)

5,730

No early VP (60.3%)

28.3%

Overall mortality



Interpretation of crude results

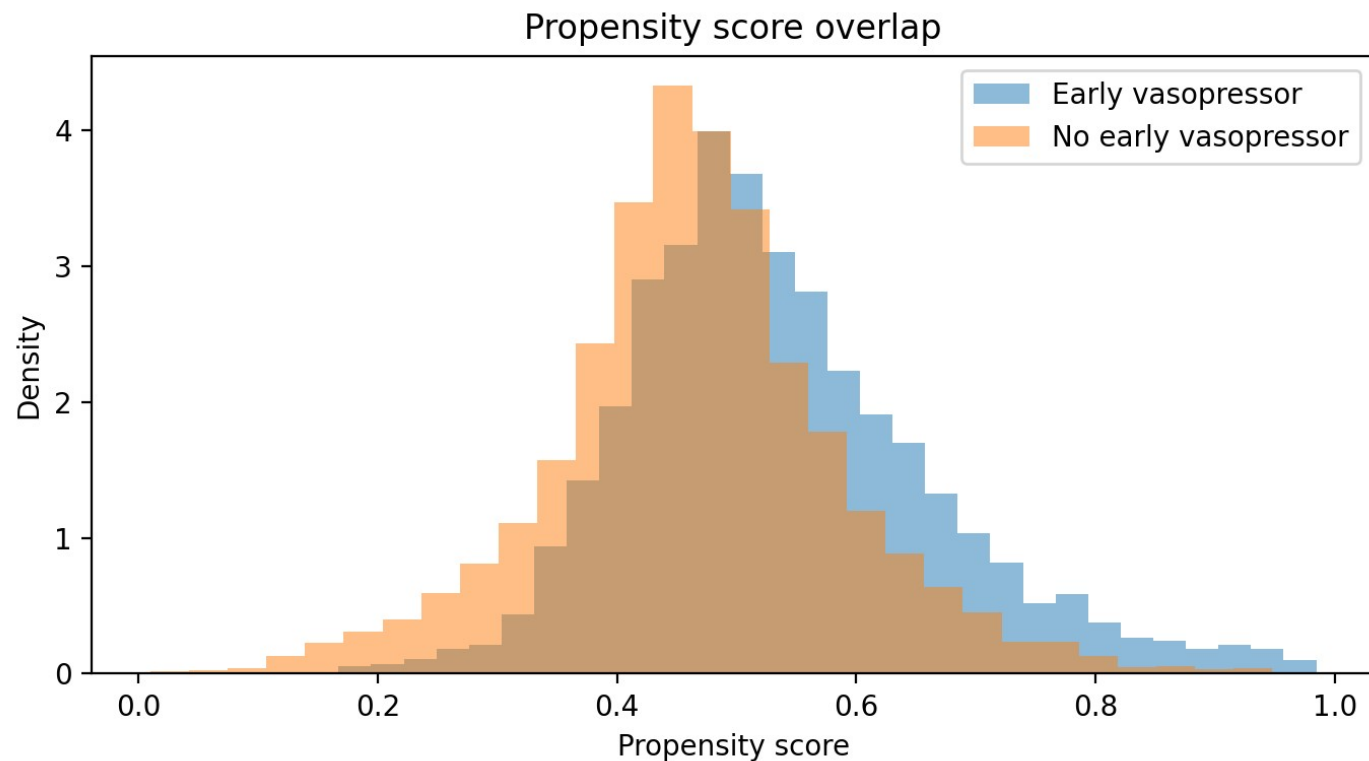
Early VP crude mortality: 32.9%

No early VP crude mortality: 25.2%

粗结果不能直接做因果解读；早期用药通常代表病情更重。

5. Propensity Score 与 Overlap

检查 treated/control 是否存在可比较区域



Key Message

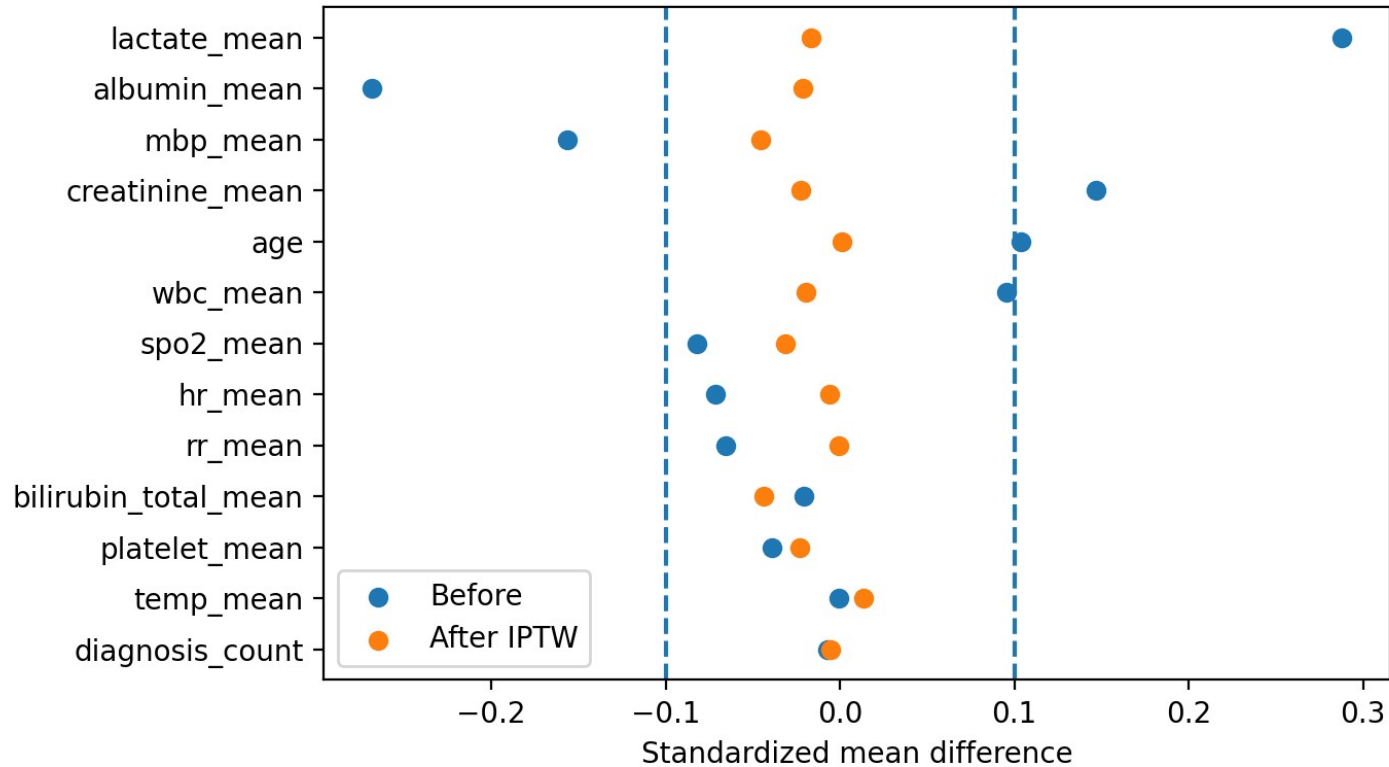
两组 propensity score 分布存在重叠，支持后续加权分析。

早期升压药组分布略偏高，符合“病情更重者更可能早期用药”的临床预期。

正式研究可进一步做 overlap trimming / positivity check。

6. Covariate Balance Diagnostics

IPTW 后 measured baseline covariates 的平衡性改善



Balance Summary

13/13 numeric covariates had $|SMD| < 0.1$ after IPTW.

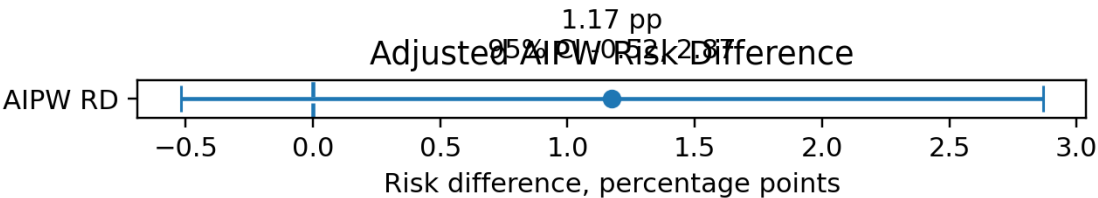
Max $|SMD|$ before weighting: 0.288.

Max $|SMD|$ after IPTW: 0.046.

Measured confounding 改善, 但 unmeasured confounding 仍需谨慎。

7. Adjusted Results

AIPW 调整后 risk difference 接近 null



Adjusted Estimate

AIPW risk difference: 1.17 percentage points.

95% CI: -0.52 to 2.87 percentage points.

置信区间跨 0；不支持从当前版本中得出明确方向性因果结论。
与粗死亡率差异相比，调整后估计明显减弱。

8. 医学解读

结论表达要强调“观察性关联”而不是“证明疗效”

For Medical Team

早期升压药是 ICU shock severity 的强信号，不能把 crude mortality 直接解释为治疗有害。

当前 adjusted RD 接近 null，提示 baseline measured covariates 调整后关联减弱。

需要复核 sepsis onset、shock 定义、升压药 itemid、液体复苏和 SOFA/Charlson。

For Boss / Business Audience

项目展示完整 RWE pipeline：cohort design → data extraction → TFL → causal adjustment → medical interpretation。

结果可用于内部方法展示和实习 / 简历项目，不应作为 clinical guidance。

下一步可升级为 protocol/SAP + sensitivity analyses + reproducible report。

9. 局限性与风险控制

把方法学风险讲清楚，反而更贴近 RWE 工作要求

Current Limitations

Sepsis 使用 ICD proxy；正式研究应验证 Sepsis-3 derived onset。

Time zero 当前为 ICU admission；可能与真实 sepsis onset 有偏差。

仍存在 residual/unmeasured confounding。

CSV malformed rows 的读取容错适合 portfolio；正式研究需源文件完整性核验。

How We Control Risk

使用 6h landmark/grace period 降低 immortal time bias。

仅使用 baseline/first-6h covariates 做 confounding adjustment。

用 PS overlap、SMD/love plot 做诊断。

报告中避免 “proved / causal proof” 等过度措辞。

10. 下一步计划

将 portfolio 项目升级为更接近正式 RWE study deliverables

Technical Next Steps

接入 mimic-code derived tables: sepsis3, SOFA, Charlson, vasopressor, ventilation 。

补充 sensitivity analyses: 0-3h/0-6h/0-12h window, early vs delayed 6-24h 。

加入 subgroup analyses: lactate/MAP/shock severity/ventilation 。

RWE Deliverables

Protocol outline: research question, estimand, cohort, endpoints, bias control 。

SAP: analysis set, missing data, PS, IPTW/AIPW, sensitivity analyses 。

TFL: cohort flow, Table 1, balance diagnostics, outcome estimates, figures 。

11. Take-home Message

这是一份更贴近 RWE & Analytics 实习岗位的项目展示

One-sentence Summary

We built a reproducible MIMIC-IV ICU RWE analysis of 9,502 adult sepsis-coded first ICU stays and showed that crude mortality differences between early and non-early vasopressor strategies were attenuated after PS/IPTW/AIPW adjustment.

Why This Fits the JD

真实世界观察性研究： cohort construction + treatment strategy + outcome analysis 。

方案 /SAP/ 报告：明确 TTE elements 、 bias control 、 statistical methods 。

数据分析： SQL/Python extraction, PS diagnostics, IPTW, AIPW, TFL outputs 。

医学沟通：主动讨论 confounding by indication 和临床定义验证。